SelectEdit: Selective Source Consistency for Inversion-Free Video Editing

Anonymous CVPR submission

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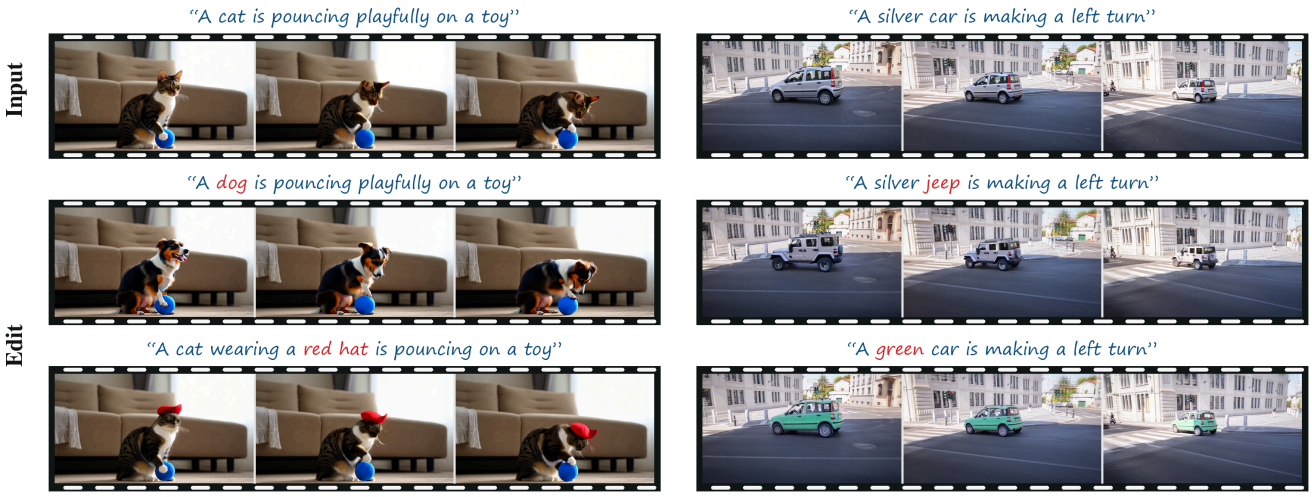


Figure 1. **Localized edits with source-scene preservation.** Cat and car examples show object replacement and local appearance changes at three time points. Each edited row corresponds to the source above; red prompt tokens identify the requested changes.

Abstract

Text-guided video editing aims to modify selected content according to a target description while preserving the surrounding scene and its motion. Existing methods face a trade-off between editing strength and source fidelity: strong source constraints can suppress the desired changes, whereas relaxed constraints can alter unrelated content and introduce temporal inconsistencies. We propose SelectEdit, a training-free, inversion-free framework that coordinates local semantic changes and detail preservation through selective source consistency. During editing, progressive trajectory regularization controls deviation from the source, while source-object attention directs semantic intervention toward the selected region. For multiscale detail reconstruction, source-motion correspondences define a shared, confidence-weighted pyramid that refines surrounding detail coherently across frames while retaining the edited coarse appearance. Experiments on all 419 FiVE-Bench tasks show that our method outperforms seven editors in structure and background preservation, edited-region text alignment, and temporal quality. It improves background PSNR by 3.88 dB over Wan-Edit and reduces temporal warp error by 5.4% relative to UniEdit-Flow.

1. Introduction

Text-guided video editing allows an object's identity, color, or material to change while the surrounding scene follows the original shot (Fig. 1). Replacing a performer with a robot, for example, requires a substantial appearance change within the performer while retaining the instrument and nearby scene. The challenge extends beyond preserving a recognizable layout: textures and boundaries outside the requested change must remain faithful to the source as they move across frames. A local edit therefore calls for selective use of source information, allowing semantic departure within the object while preserving surrounding detail along its motion.

Existing approaches control different aspects of this departure. Direct flow editing transforms source content through conditional velocity differences [20], and FlowAlign regularizes the trajectory using discrepancies between clean predictions [18]. Such source constraints reg-

ulate overall deviation, but their strength alone does not specify which changes are intended. Regional control addresses this spatial ambiguity: VideoGrain modulates cross- and self-attention to separate textual influence across regions [42], while FlowDirector spatially weights the editing flow [22]. These operations direct semantic changes, yet spatial localization by itself does not define how source textures should be recovered as the scene moves.

Temporal correspondence introduces a further distinction between consistent editing and faithful detail recovery. TokenFlow propagates edited diffusion features through inter-frame correspondences [9]; FlowDirector constrains motion through deviations from temporal appearance anchors [22]. These mechanisms operate on generated features or latent motion representations. Preserving observed scene detail additionally requires a reference for comparing the source and edited videos at corresponding surfaces. At fixed coordinates, temporal aggregation can mix different surfaces as they move. Conversely, recovering source detail without respecting the semantic edit can reintroduce content that should change. Spatial selection and motion correspondence must therefore enter the same preservation rule.

We introduce SelectEdit, a training-free, inversion-free video editor based on selective source consistency. Our central insight is to give source information complementary roles: the source object provides the spatial basis for deciding what may change, while source motion provides the correspondence basis for preserving surrounding detail. A common spatial selection guides target-semantic intervention and attenuates source-detail restoration over the same region, linking what is changed to what is preserved. Motion correspondence grounds this preservation in the observed scene as it evolves across frames.

On a progressively regularized editing trajectory, region-guided semantic modulation (RGSM) uses source-object attention to weight target-object value contributions at the current editing state. This regional intervention shapes the target prediction, and the same spatial support weights early latent transport. Motion-aligned multiscale reconstruction (MAMR) then uses source motion and correspondence confidence to define a shared pyramid for the source and decoded videos. It corrects their residual-band differences over the complementary region while retaining the edited coarsest component, recovering surrounding detail under the spatial selection established during semantic editing. The shared region determines where source detail is recovered, while motion correspondence determines the residuals to recover.

We evaluate our method against seven editors on all 419 FiVE-Bench tasks [23]. Our method achieves the best structure and background-preservation scores, edited-region text alignment, and temporal metrics in this comparison. Relative to Wan-Edit, background PSNR improves by 3.88 dB and LPIPS decreases by 27.2%. In the efficiency comparison, our pipeline takes 66.9% less time than Wan-Edit. Component ablations on the same full evaluation set show how trajectory control and regional intervention affect preservation, and how reconstruction improves perceptual and temporal quality at similar edited-region text alignment. Separate targeted comparisons further support applying regional selection to both semantic prediction and early transport, and incorporating source-motion correspondence into regional detail recovery.

Our main contributions are:

- We present SelectEdit, a training-free, inversion-free video editor that organizes local semantic change and source-detail recovery through a common spatial selection and a source-motion reference.
- We couple source-region-guided target-value modulation with region-weighted early transport, and recover complementary source detail through a shared motion-aligned multiscale pyramid.
- Evaluation on all 419 FiVE-Bench tasks shows that our method outperforms seven compared editors in structure and background preservation, edited-region text alignment, and temporal quality.

2. Related Work

Text-driven editing and source reconstruction. Pretrained text-to-image diffusion models, including latent diffusion [31], support editing through cross-attention control in Prompt-to-Prompt [12] and source-feature injection in Plug-and-Play Diffusion Features [34]. Null-text Inversion improves real-image reconstruction by optimizing unconditional embeddings while keeping model weights fixed [27]. For video, Tune-A-Video and Edit-A-Video adapt image models to an input clip [32, 38]; FateZero fuses inversion-time attentions, while Video-P2P combines video adaptation, embedding optimization, and decoupled attention control [25, 29]. AnyV2V instead combines first-frame editing with an image-to-video model and source-feature injection [19]. With rectified-flow models, RF-Solver improves numerical inversion and RF-Edit reuses source attention values [35]; DNAEdit directly aligns source noise and applies mobile velocity guidance [39]; UniEdit-Flow couples predictor–corrector inversion with regional editing guidance [15]. These approaches differ in what they adapt or reconstruct, and freezing model weights does not necessarily eliminate per-input optimization or inversion.

Spatially selective semantic control. Layered Neural Atlases, Text2LIVE, and shape-aware layered editing separate editable content from the surrounding video through explicit representations [1, 17, 21]. Ground-A-Video adds spatial grounding and preserves unedited content, while

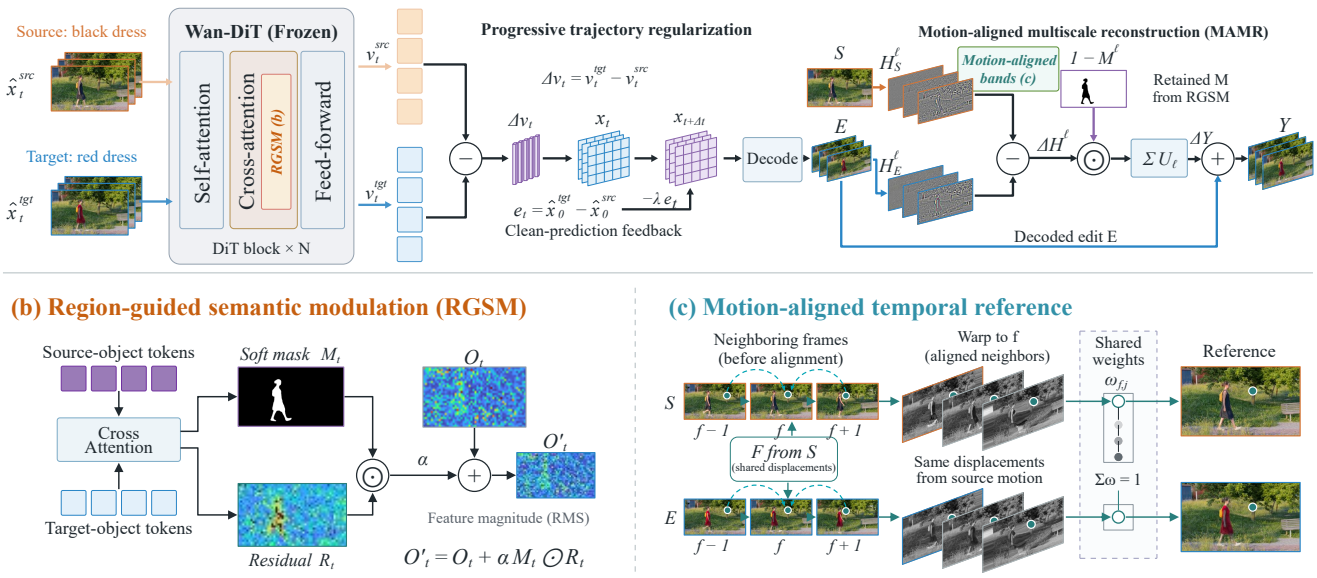


Figure 2. **Coordinating local editing and detail preservation.** A shared attention-derived region spatially weights target-token intervention and early transport along a source-consistent trajectory. After decoding, source motion defines a common pyramid for both videos. Reconstruction corrects their residual differences over the complementary region while retaining the edited coarsest component.

VideoGrain modulates cross- and self-attention for control at the class, instance, and part levels [13, 42]. VideoSwap uses sparse semantic point correspondences to accommodate shape changes [10], and ReVideo learns joint content and motion control [28]. Regional control also appears in direct flow editing: DFVEdit combines implicit-cross-attention masks with target-embedding reinforcement [5], and FlowDirector spatially weights its editing flow [22]. SelectEdit uses source-object support to weight target-object value contributions and early transport, then carries that support into complementary detail recovery.

Inversion-free editing. SDEdit edits through partial noising and denoising [26], whereas DDS optimizes editable representations using score differences and DreamMotion combines video score distillation with spatiotemporal self-similarity [11, 14]. FlowEdit directly transports source content using conditional velocity differences [20]. FlowAlign introduces clean-prediction feedback, and DRFS progressively scales the target-state shift [2, 18]. Wan-Edit extends direct flow editing to native video latents [23], while FlowAnchor stabilizes the editing signal [7]. SelectEdit adopts DRFS state shifting and FlowAlign clean-prediction feedback, coupling regional semantic intervention with source-detail reconstruction.

Temporal correspondence and multiscale preservation. TokenFlow and VidToMe propagate or merge corresponding features [9, 24]; Pix2Video, Fairy, and RAVE share information across frames [6, 16, 37]. FLATTEN, FRESCO, and Rerender A Video exploit motion or spatial correspondences [8, 40, 41]. VideoDirector uses source-conditioned guidance, while FADE introduces frequency-aware video guidance [36, 44]. Frequency-aware image editing also exploits multiscale representations [30, 33]. Building on residual pyramids and multiresolution blending [3, 4], MAMR analyzes source and decoded edits with the same source-motion and confidence operator, restores complementary residual bands, and retains the edited coarsest component.

3. Method

3.1. Inversion-Free Video Editing

Given a source video S and descriptions $c_{\text{src}}, c_{\text{tgt}}$ of its original and desired content, we seek a local semantic change that preserves the surrounding scene and motion. SelectEdit assigns distinct roles to these inputs: target text supplies new content, source-object attention specifies its spatial support, and source motion provides the correspondence reference for detail recovery. The editing region and its complement connect semantic intervention to preservation (Fig. 2). The shared selection makes detail recovery follow the intended editing scope, while source motion ties cross-frame comparisons to corresponding scene surfaces.

We use a frozen video VAE with encoder $\mathcal{E}$ and decoder $\mathcal{D}$, and a frozen velocity predictor v_θ. Rectified-flow time

$t \in [0, 1]$ decreases toward the clean endpoint $t = 0$. Initialized at $x_0^{\text{src}} = \mathcal{E}(S)$ at the selected starting time, the editing latent x_t evolves by direct source-to-target transport [20]:

$$\Delta v_t = v_t^{\text{tgt}} - v_t^{\text{src}}, \qquad x_{t+\Delta t} = x_t + \Delta t\, \Delta v_t, \tag{1}$$

where $\Delta t < 0$ advances sampling, and $v_t^{\text{src}}, v_t^{\text{tgt}}$ are source-conditioned and target-guided predictions at coupled noisy states. Decoding the endpoint gives $E = \mathcal{D}(x_0)$. We regularize this trajectory and give its semantic intervention and subsequent detail recovery a common spatial basis.

3.2. Progressive Trajectory Regularization

To establish a stable editing trajectory for localized semantic intervention, we anchor the source query to the observed video and progressively incorporate the accumulated edit into the target query through state shifting [2]. This controls how the evolving appearance influences subsequent velocity predictions. At time t, a shared noise draw $\epsilon \sim \mathcal{N}(0, I)$ defines

$$\hat{x}_t^{\text{src}} = (1 - t)x_0^{\text{src}} + t\epsilon, \tag{2}$$

and the noisy current edit

$$\bar{x}_t = (1 - t)x_t + t\epsilon. \tag{3}$$

Progressive state shifting (PSS) compensates the attenuated edit displacement in this query:

$$\hat{x}_t^{\text{tgt}} = \bar{x}_t + \kappa_t(x_t - x_0^{\text{src}}), \tag{4}$$

with $\kappa_{t_i} = \eta(i/N)t_i$ for retained steps $i = 0, \ldots, N - 1$. Here η controls compensation strength. The compensation relative to t_i grows as sampling proceeds, allowing accumulated changes to contribute more strongly to later queries. We evaluate v_t^{src} under c_{src} at $\hat{x}_t^{\text{src}}$. At $\hat{x}_t^{\text{tgt}}$, v_t^{tgt} extrapolates from the source-conditioned to the target-conditioned prediction with guidance scale w.

To keep this trajectory tied to the source as target content develops, we regulate the discrepancy between the predicted clean states through clean-prediction feedback. The paired estimates are

$$\hat{x}_0^{\text{src}} = \hat{x}_t^{\text{src}} - tv_t^{\text{src}}, \quad \hat{x}_0^{\text{tgt}} = \hat{x}_t^{\text{tgt}} - tv_t^{\text{tgt}}. \tag{5}$$

Their discrepancy $e_t = \hat{x}_0^{\text{tgt}} - \hat{x}_0^{\text{src}}$ corrects the update:

$$x_{t+\Delta t} = x_t + \Delta t\, \Delta v_t - \lambda e_t, \tag{6}$$

where $\lambda > 0$ weights the feedback. State shifting controls the evaluation point, while feedback corrects the predicted content discrepancy (Fig. 3). Together, these operations regulate the evolving edit against a global source reference. Their coefficients control how the trajectory departs from the source, but do not specify where target semantics should act. We therefore introduce the editing region into both the target prediction and its transport, giving the update an explicit spatial preference for the requested change.

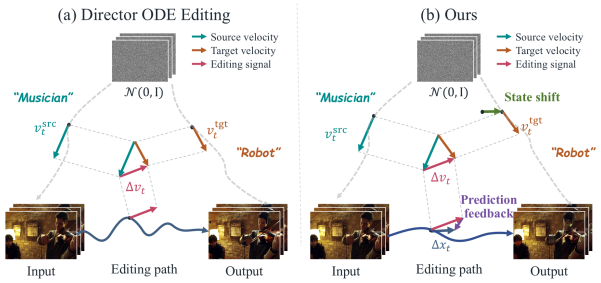


Figure 3. **State shifting and clean-prediction feedback.** Green shifts the target evaluation state; purple corrects the latent update using the clean-prediction discrepancy. Matched frames show the source and musician-to-robot edits by Wan-Edit (left) and our method (right).

3.3. Region-Guided Semantic Modulation

Local editing requires a spatial reference that remains useful as the object's appearance changes. A source object provides this reference for the requested intervention. RGSM separates spatial guidance from semantic content: source-object attention localizes the intervention, while target-object values introduce the desired appearance. Both are evaluated at $\hat{x}_t^{\text{tgt}}$, keeping localization and semantic intervention aligned with the evolving edit.

Let M_t denote the soft source-object region, and let R_t denote the attention-weighted target-object value contribution in the layer output space, including its target-attention gain. RGSM introduces the regional semantic residual

$$O_t' = O_t + \alpha M_t \odot R_t, \tag{7}$$

where O_t, O_t' are the original and modified layer outputs, α controls intervention strength, and $\odot$ is elementwise multiplication. VideoGrain modulates cross- and self-attention weights to organize regional interactions [42]; here, the added quantity explicitly carries target-object values selected by source-object support. This separates the origin of the spatial guidance from the content introduced at that location.

The same regional preference governs how this intervention advances the latent. During the first K retained steps, the corresponding latent region m_t weights the transport term in Eq. (6) as $\Delta t\, m_t \odot \Delta v_t$, while feedback remains global. The residual in Eq. (7) influences the target-conditioned prediction before the velocity difference is formed; the regional transport then weights how that difference advances the latent. Applying the same spatial preference at both points connects the formation of target content to its contribution to the update, while global feedback continues to regulate source deviation. After this window, ordinary attention and unmasked transport resume for refinement with full temporal context.

This selection also defines the scope of source-detail recovery. Its complement identifies where source residuals can be reintroduced, linking preservation to the region used for semantic intervention. The remaining requirement is to compare those residuals along the scene's motion, which we express through a shared reconstruction operator.

3.4. Motion-Aligned Multiscale Reconstruction

Regional semantic control specifies where to edit, but recovering surrounding detail requires knowing which scene content corresponds across frames. At a fixed pixel, adjacent frames may depict different surfaces; temporal averaging at that location can therefore mix unrelated detail. MAMR defines source–edit residuals through a common motion reference and corrects them under the spatial selection established by RGSM. This gives detail recovery both a semantic boundary and a temporal correspondence basis.

Let F collect source optical flows and their confidence maps. For either $X = S$ or $X = E$, the motion-aligned temporal operator is

$$(\mathcal{T}_F X)_f = \sum_{j=-r}^{r} \omega_{f,j} \odot \mathcal{W}(X_{f+j}, D_{f,j}), \tag{8}$$

where f indexes frames, r is the temporal radius, and $\mathcal{W}$ samples neighboring frame $f+j$ using the source displacement $D_{f,j}$. Normalized weights $\omega_{f,j}$ combine temporal proximity and correspondence confidence. The same displacements and weights are used for both videos, so uncertain matches contribute less to their common reference. Source motion anchors this reference to the observed scene whose surrounding detail should be retained. Reusing it for the edited video keeps the comparison tied to the same scene correspondences even as the requested appearance changes.

We incorporate this operator into every reduction of a multiscale pyramid [4]. At scale ℓ, $\mathcal{P}_{F^\ell}$ combines motion-aligned temporal aggregation with spatial reduction. Starting from $G_X^0 = X$, define

$$G_X^{\ell+1} = \mathcal{P}_{F^\ell}(G_X^\ell), \quad H_X^\ell = G_X^\ell - \mathcal{U}_\ell G_X^{\ell+1}, \tag{9}$$

where $\mathcal{U}_\ell$ upsamples to level ℓ. The shared reduction makes $H_S^\ell - H_E^\ell$ a content difference measured under identical motion neighborhoods, confidence weights, and scales. Separate motion references would also introduce differences in the neighborhoods used to define the bands, mixing content changes with changes in the comparison itself. At each scale, the band measures what remains after reduction to a motion-aligned coarser representation, capturing spatial detail and temporal variation relative to moving scene content. Temporal correspondence therefore determines the residual to recover throughout the pyramid, rather than acting solely as a smoothing operation on the final output. This differs from per-frame spatial fusion and from FlowDirector's latent motion correction around temporal appearance anchors [22]: MAMR reconstructs decoded detail using explicit source correspondences at every scale.

Let M^ℓ be the semantic editing region at scale ℓ. The corrected band is

$$\widehat{H}^\ell = H_E^\ell + (1 - M^\ell) \odot (H_S^\ell - H_E^\ell). \tag{10}$$

Reconstruction starts from the edited coarsest component and successively adds the corrected bands, recovering detail at multiple spatial scales around the edited coarse appearance. In Eq. (10), M^ℓ determines where recovery is attenuated, while $H_S^\ell - H_E^\ell$ specifies what is corrected under the shared motion reference. Larger editing weights reduce the source correction; smaller weights favor source-detail recovery. The soft weighting allows this balance to vary across the editing boundary.

The same regional preference thus has complementary roles in RGSM and MAMR: it weights the introduction of target content during editing and attenuates the return of source residuals during reconstruction. The editing region determines the strength of source recovery, the shared motion reference determines its residual content, and the edited coarsest component anchors the reconstruction. These roles connect local semantic intervention to surrounding detail preservation. The supplement provides the region, confidence, and reconstruction details.

4. Experiments

We evaluate whether selective source consistency enables localized semantic changes while preserving scene detail and motion. Comparisons on FiVE-Bench establish the overall editing quality, followed by component ablations and an analysis of inference cost.

4.1. Experimental Setup

Dataset. We evaluate on 419 editing tasks spanning six categories of FiVE-Bench [23], covering object replacement, color and material changes, object addition, and object removal. The main quantitative comparisons and component-removal ablations use this full evaluation set with matched source and target prompts. Targeted mechanism comparisons additionally use a stratified subset of 84 tasks.

Baselines. We compare with seven editors: TokenFlow [9], DNAEdit [39], DFVEdit [5], Wan-Edit [23], FlowDirector [22], UniEdit-Flow [15], and RF-Solver-Edit [35].

Metrics. We assess text alignment using global CLIP similarity to the target description and $\text{CLIP}_{\text{edit}}$ between the

Table 1. Comparison on 419 FiVE-Bench edits, macro-averaged over six categories. $\text{CLIP}_{\text{edit}}$: edited-region text alignment; Fidelity: motion fidelity; Warp: temporal warp error. Bold and underlined values indicate the best and second-best results.

	Structure and background preservation					Text alignment		Temporal quality	
Method	$\text{Dist.}_{\times 10^3} \downarrow$	PSNR ↑	$\text{LPIPS}_{\times 10^3} \downarrow$	$\text{MSE}_{\times 10^4} \downarrow$	$\text{SSIM}_{\times 10^2} \uparrow$	CLIP ↑	$\text{CLIP}_{\text{edit}} \uparrow$	$\text{Fidelity}_{\times 10^2} \uparrow$	$\text{Warp}_{\times 10^3} \downarrow$
TokenFlow [9]	21.94	20.73	157.68	98.45	72.67	<u>26.57</u>	19.91	<u>89.33</u>	31.9
DNAEdit [39]	30.78	20.50	173.29	107.91	70.21	26.43	20.24	82.40	25.6
DFVEdit [5]	13.69	24.24	98.38	43.63	80.99	25.87	19.98	89.15	26.3
Wan-Edit [23]	<u>11.53</u>	<u>25.24</u>	<u>91.52</u>	<u>36.10</u>	<u>82.87</u>	26.39	20.23	89.29	24.8
FlowDirector [22]	19.90	22.95	100.13	62.20	82.63	**26.84**	20.38	88.02	24.3
UniEdit-Flow [15]	17.66	24.50	128.86	46.26	79.21	25.89	<u>20.50</u>	88.09	<u>22.4</u>
RF-Solver-Edit [35]	59.88	18.44	280.39	205.82	62.14	25.33	19.65	69.89	23.0
Ours	**6.07**	**29.12**	**66.66**	**16.36**	**88.80**	26.37	**20.78**	**90.41**	**21.2**

edited region and the target object phrase [23]. Structure distance measures structural preservation, while PSNR, LPIPS [43], MSE, and SSIM assess background fidelity outside the annotated editing region. Following FiVE-Bench [23], motion fidelity measures the similarity of long-term point trajectories between the source and edited videos. Temporal warp error measures differences between motion-aligned adjacent frames.

Implementation details. We use the frozen Wan2.1-T2V-1.3B backbone to edit 41-frame clips at 832×480. The sampling schedule has 50 steps with editing strength 0.8 and guidance scale $w = 19.5$. We set $\eta = 1$ for PSS and $\lambda = 0.001$ for clean-prediction feedback. RGSM uses residual strength $\alpha = 0.15$ during the first $K = 10$ retained steps, and MAMR uses a four-level pyramid. Additional settings, category-wise results, and complete ablation metrics are provided in the supplement.

4.2. Main Results

Quantitative comparison. Our method improves scene preservation while achieving the highest edited-region text alignment among the compared methods (Table 1). Against Wan-Edit, the strongest baseline on all five preservation metrics, PSNR increases from 25.24 to 29.12 dB and LPIPS decreases by 27.2%. Structure distance, MSE, and SSIM also improve. These gains accompany a $\text{CLIP}_{\text{edit}}$ score of 20.78, exceeding the strongest baseline value of 20.50. Global CLIP similarity is 26.37, below FlowDirector's 26.84. The advantage in text alignment is specific to the edited region.

The preservation gains extend to temporal quality. Our method achieves the highest motion fidelity, 0.9041, and the lowest warp error, 0.0212. The latter is 5.4% below UniEdit-Flow, the strongest baseline on this measure.

Qualitative comparison. Figure 4 illustrates object replacement, recoloring, and material editing. The violinist-to-robot edit requires changing the performer while retaining the instrument and the surrounding scene. TokenFlow and Wan-Edit largely retain the original performer, whereas our result introduces a robotic appearance while preserving the violin, seated guitarist, and stone-walled room. For the uniform edit, our result changes the clothing to yellow while retaining the catcher and field layout. The metallic-hiker example presents a similar challenge at a smaller spatial scale: most baselines retain much of the hiker's original appearance, while RF-Solver-Edit introduces pronounced background artifacts. Our result gives the hiker a metallic surface while preserving the rocky surroundings.

Table 2. **User study.** Preference rates (%) for our method against seven baselines. Edit: edit alignment; Pres.: content preservation; Temp.: temporal consistency; Overall: overall preference.

Method	Edit	Pres.	Temp.	Overall
TokenFlow [9]	83.3	89.6	87.5	88.5
DNAEdit [39]	75.0	86.5	77.1	77.1
DFVEdit [5]	79.2	80.2	69.8	78.1
Wan-Edit [23]	64.6	72.9	63.5	64.6
FlowDirector [22]	64.6	80.2	80.2	74.0
UniEdit-Flow [15]	81.2	85.4	81.2	83.3
RF-Solver-Edit [35]	84.4	93.8	92.7	88.5

User study. We conduct a user study with 32 participants, each evaluating 21 randomly sampled video pairs, yielding 672 pairwise comparisons. For each pair, participants choose between our result and the baseline result on four criteria: edit alignment, content preservation, temporal consistency, and overall preference. Table 2 shows that participants prefer our method over all seven baselines across all four criteria, with overall preference reaching 88.5%. Against every baseline, content preservation receives the highest preference rate among the four criteria, consistent with the preservation gains in the quantitative evaluation.

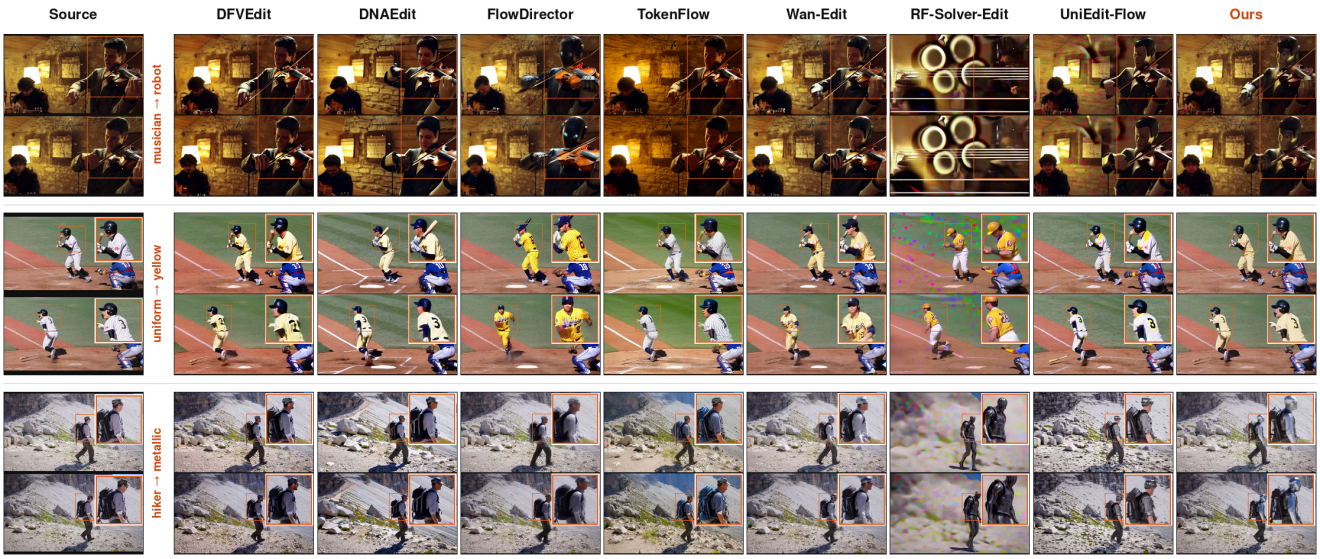


Figure 4. **Qualitative comparison on three localized edits.** Comparison with seven editors on violinist-to-robot, yellow-uniform, and metallic-hiker edits (top to bottom), with two matched time points per case. Outlines mark regions of interest; insets use identical crops across methods.

Table 3. Component removal ablation on 419 FiVE-Bench edits, macro-averaged over six categories. Each w/o row disables one component of the full pipeline. Bold and underlined values indicate the best and second-best results.

Method	PSNR ↑	LPIPS$_{\times 10^3}$ ↓	CLIP$_{\text{edit}}$ ↑	Fidelity$_{\times 10^2}$ ↑
w/o PSS	28.03	71.36	**21.05**	89.98
w/o RGSM	<u>28.87</u>	<u>68.02</u>	20.74	<u>90.31</u>
w/o MAMR	28.08	75.42	20.77	89.79
Ours	**29.12**	**66.66**	<u>20.78</u>	**90.41**

4.3. Ablation Studies

We remove PSS, RGSM, and MAMR individually from the complete method on all 419 tasks. Table 3 summarizes the resulting changes in preservation, text alignment, and motion fidelity.

PSS controls how accumulated edit displacement enters the target query, while RGSM uses source-object attention to spatially focus early semantic intervention. Removing PSS lowers PSNR from 29.12 to 28.03 dB, while increasing CLIP$_{\text{edit}}$ from 20.78 to 21.05. Disabling RGSM's target-token intervention and regional transport weighting, while retaining the source-attention region for MAMR, lowers PSNR from 29.12 to 28.87 dB and increases LPIPS from 0.06666 to 0.06802, with similar CLIP$_{\text{edit}}$ scores (20.78 versus 20.74). The bear-cap example in Fig. 5 illustrates these distinct effects. Without PSS, the cap is more prominent, but the bear's muzzle also shifts; the complete method adds the cap while keeping the muzzle closer to the source contour. With or without RGSM, the edit produces a blue cap, but RGSM gives it a smoother surface and outline. The residual visualization shows how this semantic intervention is distributed spatially.

Following semantic editing, MAMR improves detail preservation and temporal coherence together. Compared with the decoded output without reconstruction, it raises PSNR by 1.04 dB and reduces LPIPS by 11.6% and warp error by 9.5%, while CLIP$_{\text{edit}}$ remains nearly unchanged (20.77 versus 20.78). Figure 6 shows how these gains appear in the train-to-jeep edit: the reconstructed rails and sleepers retain distinct textures across consecutive frames. Within the displayed three-frame window, mean local RGB warp MAE over the tracked background region falls from 10.99 to 6.15 under shared source flow. The agreement between recovered detail and lower temporal error supports MAMR's role in preserving the surrounding scene after semantic editing.

To examine how motion correspondence contributes to detail recovery under the shared regional selection, we further compare MAMR with a spatial-only reconstruction that omits motion-aligned temporal aggregation on a stratified 84-task subset. MAMR raises PSNR from 28.23 to 29.00 dB, reduces LPIPS by 8.4%, and lowers warp error from 0.0225 to 0.0203. On the same subset, we examine whether applying this regional selection to both semantic prediction and early transport improves upon restricting transport alone. Compared with applying regional selection to early transport alone, full RGSM further improves target-semantic alignment while maintaining background fidelity. This supports applying the shared regional selection to se-

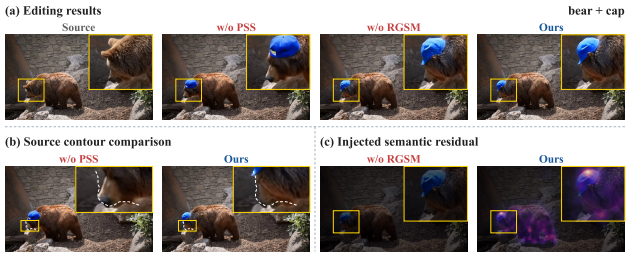


Figure 5. **Structural preservation and local semantic control.** (a) Editing results and cap details. (b) Source muzzle contours (white dashed) show the effect of PSS. (c) Relative L_2 residual magnitudes show the spatial distribution of RGSM's intervention.

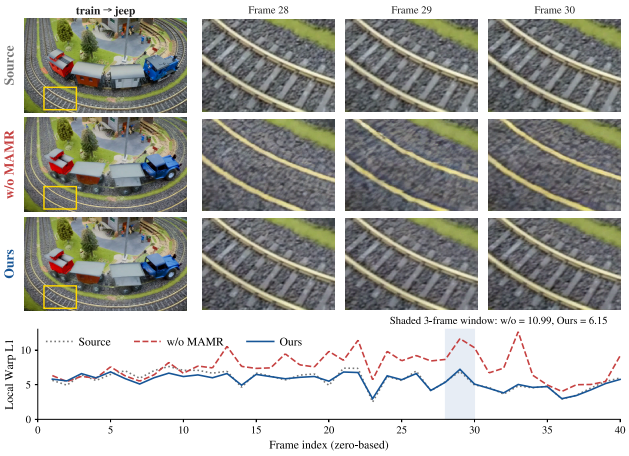


Figure 6. **Detail recovery with MAMR.** Consecutive crops show rail and sleeper textures. Curves report local RGB warp MAE under shared source flow (0–255 units); shading marks the displayed three-frame window.

mantic prediction as well as early transport. Detailed configurations and results are provided in the supplement.

4.4. Efficiency Analysis

We measure inference time and peak GPU memory on six tasks using a single NVIDIA RTX PRO 6000 Blackwell GPU. Clips have 41 frames at 832×480 and 16 fps. Each editor uses its native backbone and inference settings with offloading disabled. Timing covers the editing pipeline, including any required inversion, encoding, and decoding; model loading and file I/O are excluded.

Our complete pipeline takes 76.34 s with 23.89 GiB peak memory, including source-flow estimation and MAMR (Fig. 7). Compared with Wan-Edit (230.62 s), this reduces runtime by 66.9% at similar memory cost (24.09 GiB). UniEdit-Flow is faster at 65.67 s, while DFVEdit uses less memory at 17.93 GiB but takes 120.17 s. Our method thus combines the preservation gains in Table 1 with a shorter runtime than Wan-Edit and DFVEdit.

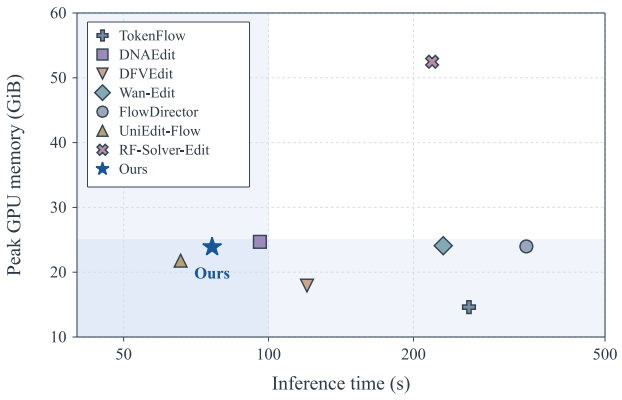


Figure 7. **Editing efficiency.** Mean inference time and maximum allocated GPU memory over 18 measured runs per method. The time axis is logarithmic; blue bands mark ≤ 100 s and ≤ 25 GiB.

4.5. Limitations

While SelectEdit effectively balances localized semantic editing with source preservation, large-scale background transformations remain challenging. For example, when converting a forested scene into a desert or a snow-covered landscape, the requested changes may affect only parts of the background, leaving some original vegetation unchanged, as illustrated in the supplementary material. To achieve more complete scene transformations, adapting source preservation to the spatial extent and semantic demands of each edit is a promising direction for future work.

5. Conclusion

We presented a training-free, inversion-free video editor that coordinates local semantic control and source-detail preservation. A shared attention-derived region spatially weights target-token intervention along a source-consistent trajectory and guides detail recovery after decoding. Source motion defines a common multiscale reference for this recovery: reconstruction corrects residual differences while retaining the edited coarsest component. The complete method improves structure, background preservation, $\text{CLIP}_{\text{edit}}$, and temporal metrics over seven compared editors on 419 FiVE-Bench edits. Matched ablations support regional control for source preservation and reconstruction for detail recovery and temporal consistency. These results support selective source consistency as a way to coordinate semantic change and scene fidelity in direct video editing.

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